

Synthetic methods note (offline teaching fixture)

Page 1. Setup and limits

A sensor records three equally spaced observations: 2, 8, and 14.

The full-window mean gives each observation weight $1/3$.

The recent-window mean keeps only the last two observations.

This is a deterministic teaching example, not an empirical benchmark.

No general claim about forecasting accuracy is tested here.

Page 2. Comparing the two outputs

Full-window weights: $(1/3, 1/3, 1/3)$. Output: 8.

Recent-window weights: $(0, 1/2, 1/2)$. Output: 11.

The output increases by 3 for this input.

The recent-window method always yields a larger output.

The last sentence is a draft claim, not a proved theorem.

Readers should distinguish this worked example from a universal claim.